1. Clinical Question & Core Hypothesis

Clinical Question

From the Ontologist synthesis, some HR+/HER2+ tumors paradoxically fail to derive durable benefit from combined HER2/ER/CDK4/6-directed regimens despite evidence of local immune organization (TLS/TILs). Why does apparent local immune organization (TLS) sometimes not translate into effective anti-tumor activity, and can a TROP2-directed ADC exploit the resulting tumor state? 2) Tier 1 paths used: I built on CORE PATH 1 (TROP2 / datopotamab deruxtecan) because it provides an actionable non-HER2 tumor-targeting axis, and CORE PATH 2 (Interleukin-18) because IL-18 links inflammatory microenvironments and dysfunctional immune phenotypes that could modulate antibody efficacy. I excluded the ambiguous “biomarker for risk of ER-negative breast cancer” from the core claim because its identity (NCBI:12) is unclear and would dilute mechanistic clarity. 3) Tier 2 support: PD-L1/immune-checkpoint modulation, Ki-67/proliferation, leptin/metabolic cross-talk and Recurrence Score (RS) all provide plausible modifiers of immune competence, ADC sensitivity, and proliferative state that can modulate the TLS–TROP2–IL-18 axis. 4) Endpoint choice rationale: I selected a therapy-response proxy (response to HER2-directed therapy vs. addition of a TROP2-directed ADC) because the central puzzle concerns differential treatment benefit and actionable combinatorial strategies (trastuzumab +/- Dato-DXd) rather than long-term survival metrics. 5) Deliberate exclusions: I kept the hypothesis tightly focused on one imaging feature (TLS-like aggregates) and two anchor entities (TROP2, Interleukin-18); other candidates (leptin, Ki-67, PD-L1, RS) are listed as secondary_modifiers for follow-up but not included in the core mechanistic sentence to preserve testability and avoid overfitting?

Working Hypothesis

In HR+/HER2+ breast cancers, TLS-like lymphoid aggregates co-occurring with high TROP2 expression and elevated IL-18 define a tumor-immune microenvironment in which anti-HER2 therapy yields reduced therapy_response due to IL-18–driven immunosuppression and TLS functional impairment. In contrast, a TROP2-directed ADC (e.g., datopotamab deruxtecan) delivers cytotoxic payload directly to TROP2-expressing tumor cells and thus achieves superior therapy_response in the same TLS-high/IL-18-high/TROP2-high subgroup, effectively bypassing reliance on immune effector functions such as ADCC. This differential response is modulated by secondary modifiers (leptin signaling, Ki-67 proliferation, PD-L1 expression), which may amplify or mitigate the observed effects in a given tumor context. The primary endpoint is therapy_response; the imaging feature is TLS-like lymphoid aggregates; anchor genes are TROP2 and IL-18; an ambiguous ER-negative risk biomarker is excluded to maintain mechanistic clarity. Testable predictions include: (i) TLS-high/IL-18-high/TROP2-high tumors show limited benefit from HER2-directed therapy; (ii) Dato-DXd yields higher response rates in the same subgroup; (iii) perturbing IL-18 or TROP2 expression in preclinical models alters these responses; and (iv) adding a TROP2-directed ADC to HER2-directed therapy yields additive or synergistic benefit in TLS-wealthy microenvironments.

Element	Value
Imaging feature	TLS-like lymphoid aggregates
Anchor gene(s)	TROP2, Interleukin-18
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Primary hypothesis gene(s)

TROP2: TROP2-directed ADC exploit the resulting tumor state? 2) Tier 1 paths used: I built on CORE PATH 1 (TROP2 / datopotamab deruxtecan) because it provides an actionable non-HER2 tumor-targeting axis, and CORE PATH 2 (Interleukin-18) because IL-18 links inflammatory microenvironments and dysfunctional immune phenotypes that could modulate antibody efficacy.

Interleukin-18: Interleukin-18); other candidates (leptin, Ki-67, PD-L1, RS) are listed as secondary_modifiers for follow-up but not included in the core mechanistic sentence to preserve testability and avoid overfitting.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	leptin	Supporting modifier from knowledge graph
Secondary	Ki-67 (IHC MIB-1)	Supporting modifier from knowledge graph
Secondary	PD-L1	Supporting modifier from knowledge graph
Secondary	Recurrence Score (RS)	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: retrospective correlative analyses of TLS density/architecture, IL-18 signaling signatures, and TROP2 expression with responses to trastuzumab-containing regimens and, where available, to Dato-DXd;

Prediction 2: in vitro assays using HR+/HER2+ cell lines with manipulated TROP2 and IL-18 levels in co-cultures with NK/immune effector cells to measure ADCC and ADC-mediated cytotoxicity;

Prediction 3: in vivo studies in humanized or TLS-mimicking mouse models to compare tumor responses to trastuzumab, Dato-DXd, or combination therapy; and

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Novel mechanistic hypothesis derived from knowledge graph analysis.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- leptin: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- Ki-67 (IHC MIB-1): Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.